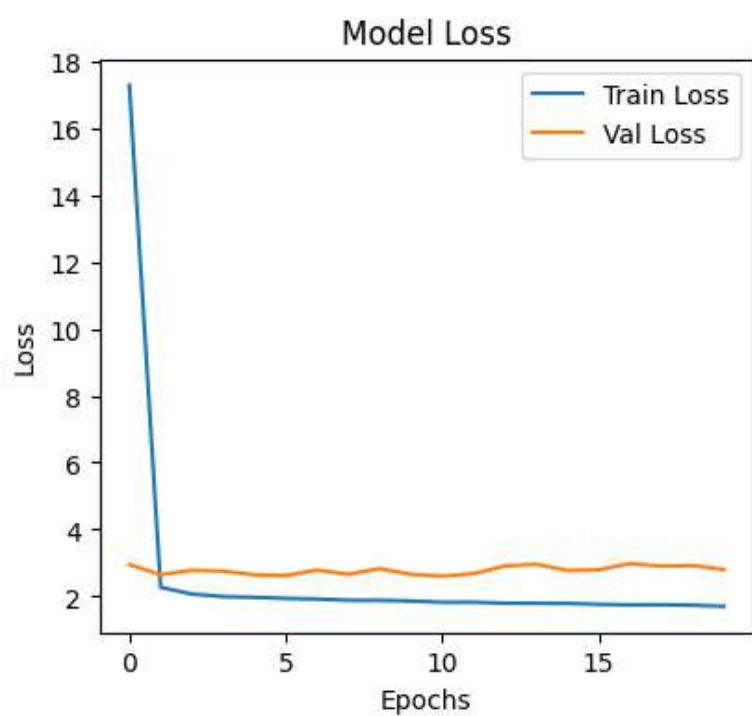
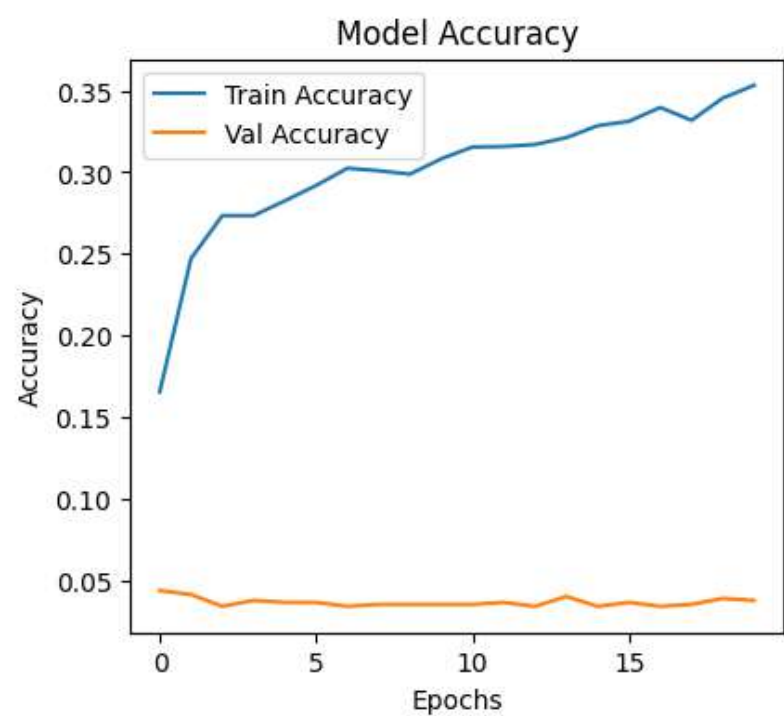
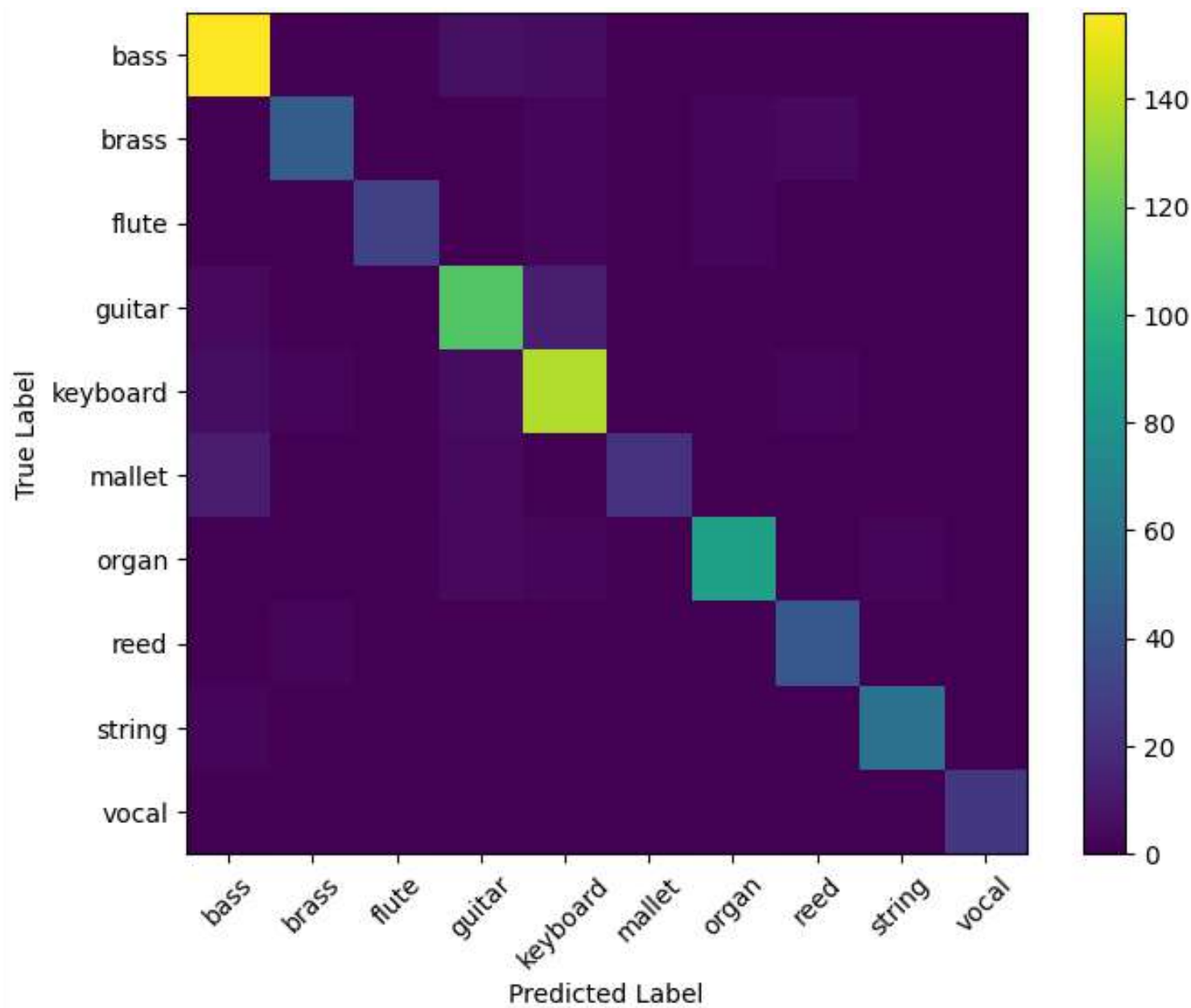


```
Epoch 1/20
/usr/local/lib/python3.12/dist-packages/keras/src/layers/core/dense.py:93: UserWarning: Do not pass an `input_shape`/`input_dim` argument to a layer. When using Sequential model
  super().__init__(activity_regularizer=activity_regularizer, **kwargs)
104/104 — 1s 3ms/step - accuracy: 0.1519 - loss: 34.6182 - val_accuracy: 0.0436 - val_loss: 2.9521
Epoch 2/20
104/104 — 0s 2ms/step - accuracy: 0.2319 - loss: 2.3637 - val_accuracy: 0.0412 - val_loss: 2.6496
Epoch 3/20
104/104 — 0s 2ms/step - accuracy: 0.2624 - loss: 2.0738 - val_accuracy: 0.0339 - val_loss: 2.7801
Epoch 4/20
104/104 — 0s 2ms/step - accuracy: 0.2614 - loss: 2.0056 - val_accuracy: 0.0375 - val_loss: 2.7560
Epoch 5/20
104/104 — 0s 2ms/step - accuracy: 0.2916 - loss: 1.9446 - val_accuracy: 0.0363 - val_loss: 2.6473
Epoch 6/20
104/104 — 0s 2ms/step - accuracy: 0.2846 - loss: 1.9411 - val_accuracy: 0.0363 - val_loss: 2.6241
Epoch 7/20
104/104 — 0s 2ms/step - accuracy: 0.3121 - loss: 1.9283 - val_accuracy: 0.0339 - val_loss: 2.7862
Epoch 8/20
104/104 — 0s 2ms/step - accuracy: 0.2962 - loss: 1.8830 - val_accuracy: 0.0351 - val_loss: 2.6636
Epoch 9/20
104/104 — 0s 2ms/step - accuracy: 0.2897 - loss: 1.9067 - val_accuracy: 0.0351 - val_loss: 2.8291
Epoch 10/20
104/104 — 0s 2ms/step - accuracy: 0.3080 - loss: 1.8639 - val_accuracy: 0.0351 - val_loss: 2.6609
Epoch 11/20
104/104 — 0s 2ms/step - accuracy: 0.3313 - loss: 1.8195 - val_accuracy: 0.0351 - val_loss: 2.6063
Epoch 12/20
104/104 — 0s 2ms/step - accuracy: 0.3211 - loss: 1.8315 - val_accuracy: 0.0363 - val_loss: 2.6797
Epoch 13/20
104/104 — 0s 2ms/step - accuracy: 0.3206 - loss: 1.7963 - val_accuracy: 0.0339 - val_loss: 2.9080
Epoch 14/20
104/104 — 0s 2ms/step - accuracy: 0.3231 - loss: 1.8031 - val_accuracy: 0.0400 - val_loss: 2.9623
Epoch 15/20
104/104 — 0s 2ms/step - accuracy: 0.3313 - loss: 1.7939 - val_accuracy: 0.0339 - val_loss: 2.7765
Epoch 16/20
104/104 — 0s 2ms/step - accuracy: 0.3375 - loss: 1.7515 - val_accuracy: 0.0363 - val_loss: 2.7944
Epoch 17/20
104/104 — 0s 2ms/step - accuracy: 0.3422 - loss: 1.7732 - val_accuracy: 0.0339 - val_loss: 2.9846
Epoch 18/20
104/104 — 0s 2ms/step - accuracy: 0.3182 - loss: 1.7713 - val_accuracy: 0.0351 - val_loss: 2.9051
Epoch 19/20
104/104 — 0s 2ms/step - accuracy: 0.3470 - loss: 1.7467 - val_accuracy: 0.0387 - val_loss: 2.9219
Epoch 20/20
```



	precision	recall	f1-score	support
bass	0.87	0.91	0.89	171
brass	0.88	0.85	0.87	54
flute	0.89	0.86	0.87	36
guitar	0.84	0.86	0.85	133
keyboard	0.84	0.89	0.86	155
mallet	1.00	0.57	0.73	40
organ	0.90	0.89	0.89	100
reed	0.86	0.91	0.89	47
string	0.94	0.94	0.94	62
vocal	1.00	0.89	0.94	28
accuracy			0.88	826
macro avg	0.90	0.86	0.87	826
weighted avg	0.88	0.88	0.87	826



```
[[156  0  1  7  5  0  1  1  0  0]
 [  0 46  0  0  2  0  2  4  0  0]
 [  0  0 31  0  3  0  2  0  0  0]
 [  4  0  0 114 13  0  1  0  1  0]
 [  6  2  1  5 138  0  1  2  0  0]
 [ 12  0  0  4  1 23  0  0  0  0]
 [  0  0  1  4  3  0 89  0  3  0]
 [  0  3  0  0  0  0  1 43  0  0]
 [  2  0  0  1  0  0  1  0 58  0]
 [  0  1  1  0  0  0  1  0  0 25]]
```